

Final/Midterm Exam
International Campus, DDS ,PharmaD
Program
Biochemisrty
February 2023 semester

Name and Surname:

Student ID Number:

Course Coordinator: Dr. Noroozi

Date of Exam: Jul 12, 2023

Duration of Exam: 60 Minutes

MCQ Question	57
--------------	----

Directions:

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil.

10	We have an effective buffer when the pH of a solution is:	
✓	1	Within approximately ± 1 pH unit of the pKa
	2	Within approximately ± 2 pH unit of the pKa
	3	Within approximately ± 1 pH unit of the Ka.
	4	Within approximately ± 2 pH unit of the Ka.
11	Which of the following electrolytes has the highest concentration in intracellular fluid (ICF) in comparison with extracellular fluid (ECF)?	
	1	Potassium ion
✓	2	Sodium ion
	3	Chloride ion
	4	Bicarbonate ion
12	Which statement is true about the buffering system?	
✗	1	They tend to resist changes in pH when large amounts of acid or base are added
✓	2	It consists of a weak acid and its conjugate base
✗	3	Buffering power for all conjugate acid-base pairs is the same.
✗	4	We need a buffer only in pathological condition
13	Which of the following monosaccharides is a universal fuel of the fetus?	
	1	Xylulose
✓	2	Glucose
	3	Fructose
	4	Arabinose
14	Which statement is true about cellulose?	
✗	1	Intestinal enzymes in humans can hydrolyze it.
✓	2	It is an important source of bulk in the diet.
✗	3	It is a heteropolysaccharide
✗	4	It is the main energy reserve in human
15	A researcher wants to make a cream for wound repair. Which of the following carbohydrates can be an appropriate candidate for this purpose?	
	1	Hyaluronic acid
✓	2	Heparin
	3	Inulin
	4	Maltose
16	Osmotic laxatives can soften your stools and make them easier to pass by increasing the amount of water in your bowels. Which of the following biomolecules can be used as a mild osmotic laxative?	
	1	Inulin
	2	Lactulose
✓	3	Amylose ✗
	4	Isomaltose ✗
17	A high concentration of xylulose was seen in an infant's urine. What is a possible diagnosis?	
	1	Pentosuria
✓	2	Diabetes mellitus
	3	Lactose intolerance
	4	Sucrose intolerance

18	Which statement is true about carbohydrate isomerism?
☒	1 Most of the naturally occurring monosaccharides are L isomers
☒	2 Functional (Aldose-ketose) isomerism has the different molecular formula
☒	3 Epimers are mirror images of each other
☒	4 Sucrose is an invert sugar
19	Which of the following carbohydrates is an important structural compound in Flavin coenzymes?
☒	1 Ribose
	2 Arabinose
	3 Xylose
	4 Maltose
20	Which of the following biomolecules can be used for the biosynthesis of anti-coagulants?
	1 Keratan sulfate
☒	2 Heparin
	3 Heparan sulfate
	4 Amylopectin
21	Which organ uses glucose as the only metabolic fuel in both the fed state and the starving state?
☒	1 Mature erythrocytes
	2 Liver ✓
	3 Brain ✗
	4 Muscle ✗
22	Which of the following statements about fatty acids is correct?
☒	1 One of them is the precursor of prostaglandins.
	2 They all contain one or more double bonds.
	3 They are constituents of sterols.
	4 They are extremely hydrophilic.
23	Which of the following fatty acid has the lowest melting point?
	1 Palmitic acid
	2 Stearic acid
	3 Arachidonic acid
☒	4 Eicosapentaenoic acid
24	Which of the following has an ether bond?
	1 Gangliosides ✗
	2 Phosphatidyl serine
☒	3 Platelet-activating factor ✓
	4 Sphingomyelin
25	Which of the following lipoproteins is rich in cholesterol?
	1 Chylomicron
	2 VLDL
☒	3 LDL
	4 HDL

26	Sphingosine is a component of all of the following, <u>except</u> :
✓ 1	Gangliosides
2	Cardiolipin
3	Sphingomyelin
4	Ceramide
27	Which of the following is an essential fatty acid?
1	Oleic acid
2	Arachidonic acid
✓ 3	Linolenic acid
4	Palmitoleic acid
28	The number of double bonds in arachidonic acid is....
1	1
2	2
3	4
4	6
29	Which of the following statement is <u>not</u> true about the K_m (Michaelis constant)?
1	Has a lower numerical value when the enzyme has high affinity for its substrate.
2	Is the substrate concentration when the reaction rate is half of the maximum rate.
3	Is a measure of the affinity of the enzyme for its substrate. ✓
✓ 4	Has a higher numerical value when the enzyme has high affinity for its substrate.
30	Which of the following is the definition of enzyme activity?
1	The amount of the enzyme that catalyzes the conversion of one micromole of substrate per second.
2	The number of molecules of substrate converted to product per enzyme molecule per second.
3	✗ The amount of the enzyme that catalyzes the conversion of one micromole of substrate per minute.
4	✗ The number of molecules of substrate converted to product per enzyme molecule per minute.
31	In the modern system of nomenclature which one of the following enzyme occupies third position?
1	Hydrolase
2	Ligase
✓ 3	Oxidoreductase
4	Transferase
32	Which of the following statement about enzymes is <u>not</u> true:
1	Enzymes accelerate reactions by lowering the activation energy ✓
✓ 2	Enzymes are proteins whose three-dimensional form is key to their function ✓
3	The enzymes change the equilibrium of the reaction ✓
4	Enzymes do not change the free energies of the reactants or products ✓
33	Which of the following is not an example for competitive inhibitor?
1	Methotrexate for dihydrofolate reductase ✓
2	Atorvastatin for HMG-CoA reductase ✓
3	Penicillin for transpeptidase ✓
✓ 4	Malonate for succinate dehydrogenase

34	Which of the following statement about isoenzymes is <u>not</u> true:
	1 Display different regulatory properties ✓
	2 Have distinct physical properties ✓
✓	3 Catalyze the different reaction
	4 May be expressed in a tissue-specific ✓
35	In allosteric Enzyme, effector binding site is known as:
✓	1 R-subunit
	2 C-subunit
	3 T-subunit
	4 M-subunit
36	Biotin act as a coenzyme for all except:
	1 Pyruvate to Oxaloacetate
	2 Acetyl CoA to Malonyl CoA
✓	3 Propionyl CoA to Methyl Malonyl CoA
	4 Glutamate to Gamma Carboxy Glutamate
37	Name the coenzyme of riboflavin (B2)?
	1 NAD or NADP
✓	2 FAD and FMN
	3 Coenzyme A ✗
	4 Thiamine pyrophosphate ✗
38	Which of the following diseases is caused by the deficiency of Niacin?
	1 Scurvy ✗
	2 Rickets ✗
✓	3 Pellagra
	4 Pernicious anemia
39	Which of the following is effective in premenstrual syndrome (PMS) and diabetes?
	1 Vitamin B12
	2 Vitamin B9
✓	3 Vitamin B6
	4 Vitamin B3
40	Which vitamin deficiency causes neural tube defects?
	1 Vitamin C ✗
	2 Vitamin B12
✗	3 Niacin
✓	4 Folic acid ✓
41	The coenzyme forms of vitamin B12 are:
	1 Hydroxycobalamin- Cyanocobalamin
	2 Adenosylcobalamin - Methylcobalamin
	3 Cyanocobalamin – Adenosylcobalamin
✓	4 Methylcobalamin - Hydroxycobalamin
42	Which of the following organs are responsible for the activation of vitamin D in the human body?
	1 Intestine and Liver
	2 Kidney and Pancreas
✓	3 Liver and Kidney
✗	4 Stomach and Brain

43	Increase in the fragility of red blood cells is caused due to the
✓ 1	Deficiency of vitamin E
2 ×	Deficiency of vitamin D
3 ×	Deficiency of vitamin C
4 ×	Deficiency of vitamin A
44	The enzyme DNA ligase....
1	Introduces super helical twists ×
✓ 2	Connects the end of two DNA chains
3	Unwinds the double helix
✓ 4	Synthesizes RNA primers
45	RNA primer is formed by the enzyme:
1	Ribonucleas
✓ 2	Primase
3	DNA polymerase I
4	DNA polymerase III
46	During replication, unwinding of double helix is initiated by
1	DNAA protein
✓ 2	DnaB protein
3	DNAC protein
✓ 4	Rep protein
47	Substitution of a thymine base by adenine in DNA is known as
1	Transposition
✓ 2	Transition
3	Transversion
4	Frameshift mutation
48	Xeroderma pigmentosum results from a defect in....
1	UVR ABC excinuclease
✓ 2	DNA polymerase I
3	DNA ligase
4	DNA polymerase III
49	Genetic code is
✓ 1	Overlapping
2	Non-overlapping
3	Not universal
4	Ambiguous
50	The enzyme amino acyl tRNA synthetase is involved in....
1	Dissociation of discharged tRNA from 80S ribosome
2	Charging of tRNA with specific amino acids
3	Termination of protein synthesis
4	Nucleophilic attack on esterified carboxyl group of peptidyl tRNA
51	In prokaryotes, chloramphenicol
1	Causes premature release of the polypeptide chain
2	Causes misreading of the mRNA
3	Depolymerizes DNA ×
4	Inhibits peptidyl transferase activity

52	✓	Actinomycin D binds to
	1	Double stranded DNA
	2	Single stranded DNA
	3	Single stranded RNA
	4	DNA-RNA hybrid
53		In eukaryotes, the 40 S pre-initiation complex contains all the following initiation factors <u>except</u> :
	1	eIF-1A
	2	eIF-2
	3	eIF-3
	4	eIF-4
54		Tetracyclines inhibit binding of amino acyl tRNAs to
	1	30 S ribosomal subunits
	2	40 S ribosomal subunits
	3	50 S ribosomal subunits
	4	60 S ribosomal subunits
55	✓	The gene of lac operon which has constitutive expression is.....
	1	I
	2	C
	3	Z
	4	P
56		cis-acting elements include:
	1	Steroid hormones
	2	Calcitriol
	3	Histones
	4	Silencers
57	✓	Which of the following bacterial operon is not controlled by attenuation?
	1	Arabinose
	2	Tryptophan
	3	Leucine
	4	Histidine